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R Notebook: Raw Data Aggregation

The Raw Data Aggregation R-Notebook aggregates the raw signal data by stimulus across respondents. This article describes how you can run this R-Notebook in iMotions and the aggregation algorithm behind.

1. Prerequisites

Supported stimulus types: all

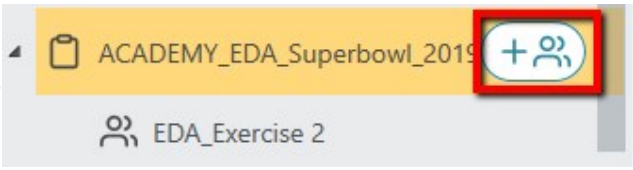
Supported sensors: all

Supported iMotions version: iMotions 8.1 and later

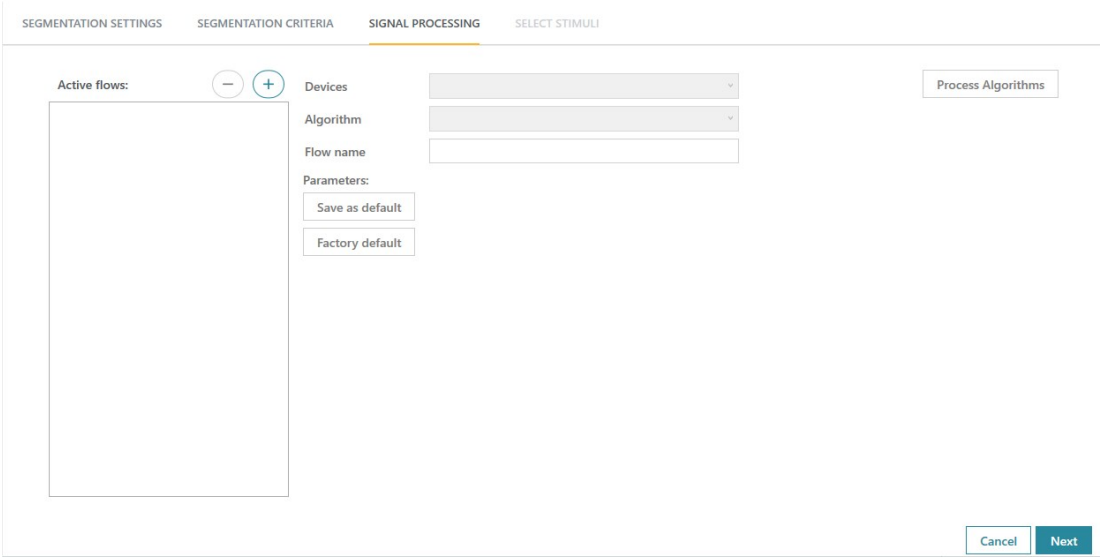
2. How to Analyze in iMotions Desktop

As for Raw Data Aggregation R-Notebook, the following steps need to be followed:

1. Create a new analysis for the study for which you wish to run the aggregation R-Notebook.



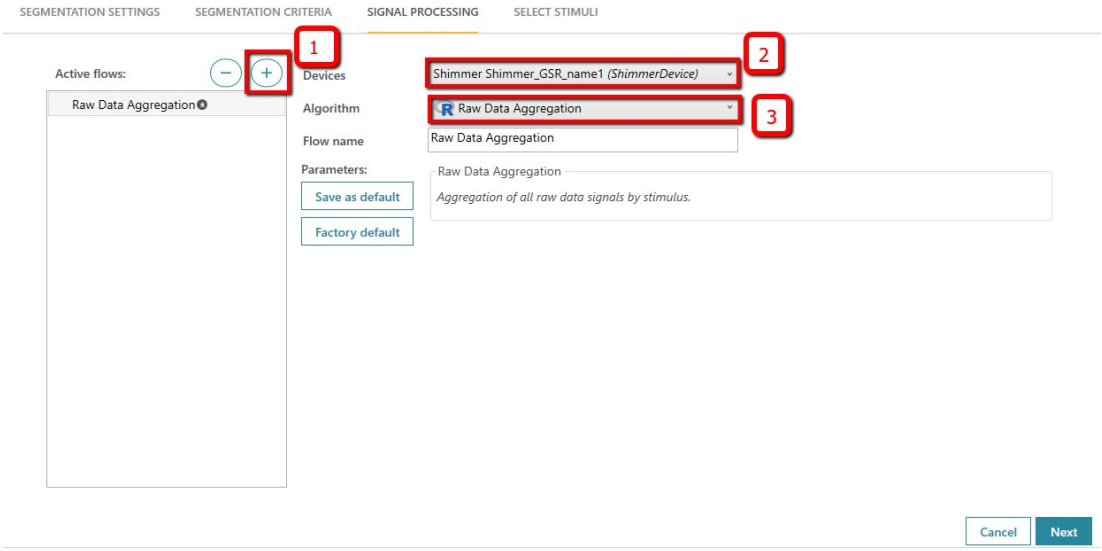
2. Advance through the steps "Segmentation settings" with your preferred selections and "Segmentation criteria" to choose respondents you want to do aggregation on until you reach the step "Signal processing".



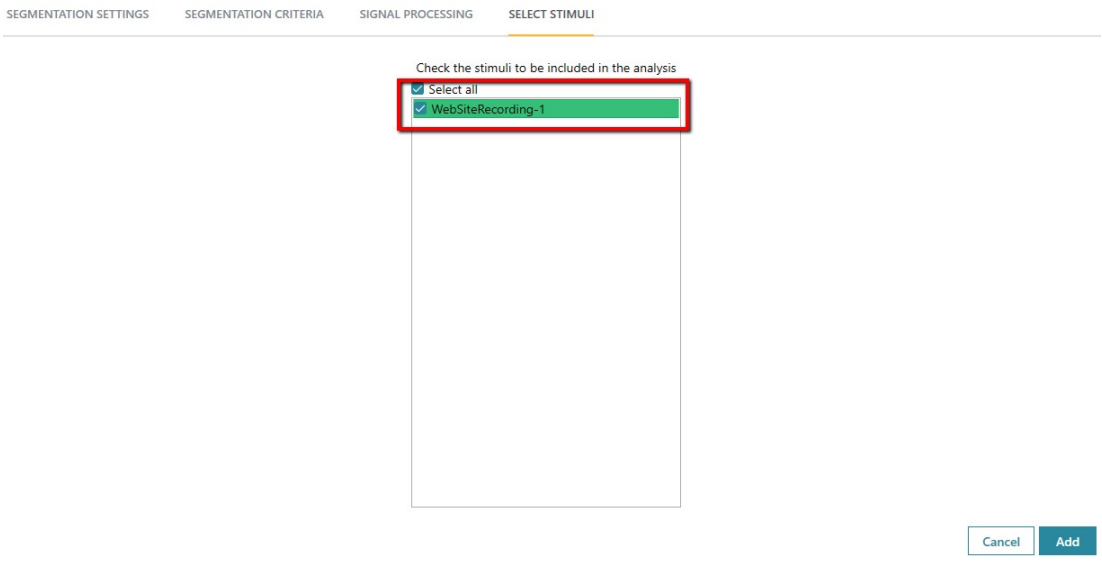
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Note: the aggregation R-Notebook will not be executed if there is only one respondent or no respondents available for aggregation.

3. By clicking "+" next to "Active flows" you can select the algorithms you want to use for signal processing. Select the sensor for which you want to apply the aggregation R-Notebook in the drop-down "Device" menu. Select the algorithms in the drop-down “Algorithm” menu.
   Note: it is possible now (from version 8.1 and onwards), to run the notebook on multiple sensors of the same kind. E.g., you can run the notebook on the second Shimmer device here by adding an additional Active flows with the device you would like to analyze on.



4. Name the analysis in the "Flow name" menu.
5. Select the stimulus to be included in the analysis. Then click the "Add" button to add the analysis and close the dialog.



3. Aggregation Algorithm Introduction

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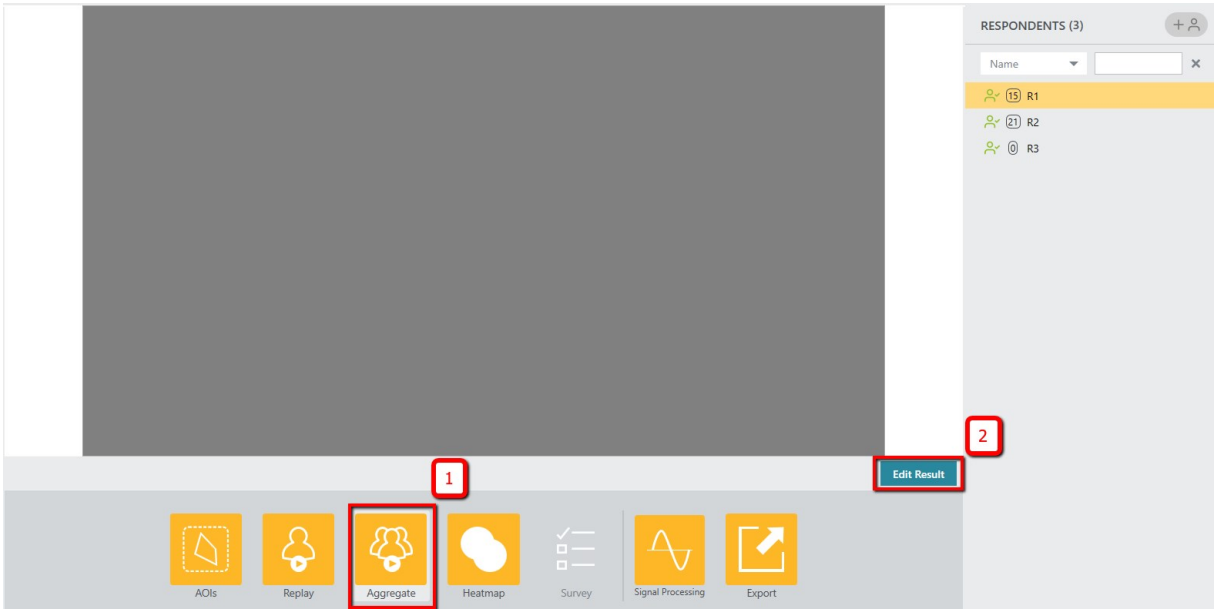
1. The R-Notebook will not be executed if there is only one respondent or no respondents available for aggregation.

2. First, the data of all available respondents of the specified sensor and stimulus are retrieved. The raw data aggregation algorithm will be run for each signal channel of the selected sensor.

3. Within each signal channel, the mean signal across all respondents of the specified stimulus are calculated and stored. The generated time-series mean signal is the aggregated signal.

4. Outputs

By clicking 'Aggregate' button at the bottom and then 'Edit Result', you can enter the aggregate replay:



You can also switch to aggregate replay in the replay window by clicking 'AGGREGATE' at the top of the replay window:

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Here in the replay window, you will see the aggregated signals of the specified stimulus and sensor. The Falloff curve shows how many respondents are available for aggregation over time.

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